

Area in Polar Coordinates

ANSWER KEY



The polar area formula

- 1 State the formula for the area enclosed by a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$.

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 d\theta.$$

- 2 Find the area enclosed by one full loop of $r = 3\cos \theta$ (i.e. $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$).

$$A = \frac{1}{2} \int_{-\pi/2}^{\pi/2} 9\cos^2 \theta d\theta = \frac{9}{2} \int_{-\pi/2}^{\pi/2} \frac{1+\cos 2\theta}{2} d\theta = \frac{9}{4} \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2} = \frac{9}{4}(\pi) = \frac{9\pi}{4}.$$

Area of a full cardioid

- 3 Find the total area enclosed by the cardioid $r = 1 + \cos \theta$.

$$A = \frac{1}{2} \int_0^{2\pi} (1 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (1 + 2\cos \theta + \cos^2 \theta) d\theta. \text{ Using } \cos^2 \theta = \frac{1+\cos 2\theta}{2}: \\ = \frac{1}{2} \left[\theta + 2\sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{2\pi} = \frac{1}{2} (2\pi + \pi) = \frac{3\pi}{2}.$$

Area between two polar curves

- 4 Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

$$A = \frac{1}{2} \int_0^{2\pi} (3^2 - 2^2) d\theta = \frac{1}{2} \int_0^{2\pi} 5 d\theta = \frac{1}{2} (5)(2\pi) = 5\pi.$$

Area of one petal of a rose curve

- 5 Find the area of one petal of $r = 2\sin(2\theta)$, using the interval $\theta \in [0, \frac{\pi}{2}]$ where one petal is traced.

$$A = \frac{1}{2} \int_0^{\pi/2} 4\sin^2(2\theta) d\theta = 2 \int_0^{\pi/2} \frac{1-\cos 4\theta}{2} d\theta = \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{\pi/2} = \frac{\pi}{2}.$$

Setting up area between curves that intersect

- 6 Set up (but do not evaluate) the integral for the area inside both $r = 2$ and $r = 2 + 2\cos \theta$, noting they intersect where $2 = 2 + 2\cos \theta$, i.e. $\theta = \pm \frac{\pi}{2}$.

By symmetry, for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ the circle $r = 2$ is the inner (smaller) boundary, while for $\theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$ the cardioid is inner. Area = $\frac{1}{2} \int_{-\pi/2}^{\pi/2} 2^2 d\theta + \frac{1}{2} \int_{\pi/2}^{3\pi/2} (2 + 2\cos \theta)^2 d\theta$.

Area of a full rose curve

- 7 Find the total area enclosed by all 4 petals of $r = 3\cos(2\theta)$, using the fact that one petal is traced for $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

Area of one petal: $\frac{1}{2} \int_{-\pi/4}^{\pi/4} 9\cos^2(2\theta) d\theta = \frac{9}{2} \int_{-\pi/4}^{\pi/4} \frac{1+\cos 4\theta}{2} d\theta = \frac{9}{4} [\theta + \frac{\sin 4\theta}{4}]_{-\pi/4}^{\pi/4} = \frac{9}{4} \cdot \frac{\pi}{2} = \frac{9\pi}{8}$. Total for all 4 petals: $4 \times \frac{9\pi}{8} = \frac{9\pi}{2}$.

Area inside a limaçon

- 8 Find the area enclosed by the limaçon $r = 3 + \cos \theta$ (a curve without an inner loop, since the constant exceeds the coefficient).

$$A = \frac{1}{2} \int_0^{2\pi} (3 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (9 + 6\cos \theta + \cos^2 \theta) d\theta = \frac{1}{2} [9\theta + 6\sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}]_0^{2\pi} = \frac{1}{2} (18\pi + \pi) = \frac{19\pi}{2}.$$

- 9 Find the area of the small inner loop of the limaçon $r = 1 + 2\cos \theta$ (constant less than coefficient, so an inner loop exists), traced for $\theta \in [\frac{2\pi}{3}, \frac{4\pi}{3}]$.

$$A = \frac{1}{2} \int_{2\pi/3}^{4\pi/3} (1 + 2\cos \theta)^2 d\theta = \frac{1}{2} \int_{2\pi/3}^{4\pi/3} (1 + 4\cos \theta + 4\cos^2 \theta) d\theta = \frac{1}{2} [3\theta + 4\sin \theta + \sin 2\theta]_{2\pi/3}^{4\pi/3}. \text{ At } \frac{4\pi}{3}: 4\pi - 2\sqrt{3} + \frac{\sqrt{3}}{2}. \text{ At } \frac{2\pi}{3}: 2\pi + 2\sqrt{3} - \frac{\sqrt{3}}{2}. \text{ Difference: } 2\pi - 3\sqrt{3}. A = \frac{1}{2} (2\pi - 3\sqrt{3}) = \pi - \frac{3\sqrt{3}}{2} \approx 0.54.$$

Setting up area problems (AP-style, no evaluation)

- 10 Set up (but do not evaluate) the integral for the area inside the rose $r = 2\sin(3\theta)$ that lies in the first quadrant only, for one petal traced on $\theta \in [0, \frac{\pi}{3}]$.

$$A = \frac{1}{2} \int_0^{\pi/3} [2\sin(3\theta)]^2 d\theta = \frac{1}{2} \int_0^{\pi/3} 4\sin^2(3\theta) d\theta.$$

- 11 Set up (but do not evaluate) the integral for the area between the circle $r = 5$ and the cardioid $r = 5 - 5\sin \theta$ (inside the circle, outside the cardioid), for $\theta \in [0, \pi]$.

$$\text{Area} = \frac{1}{2} \int_0^{\pi} [5^2 - (5 - 5\sin \theta)^2] d\theta.$$

Finding intersection points of polar curves

- 12 Find all points of intersection of $r = 2\cos \theta$ and $r = 2\sin \theta$ (both circles of radius 1) by setting the equations equal.

$2\cos \theta = 2\sin \theta \Rightarrow \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$. At $\theta = \frac{\pi}{4}$: $r = 2\cos(\frac{\pi}{4}) = \sqrt{2}$. At $\theta = \frac{5\pi}{4}$: $r = 2\cos(\frac{5\pi}{4}) = -\sqrt{2}$ (a negative r , representing the same point reached differently). Both curves also pass through the pole ($r = 0$) at different angles, which must be checked separately.

- 13 Explain why checking intersections of two polar curves algebraically (setting $f(\theta) = g(\theta)$) can MISS the pole as an intersection point, and why it must be checked separately.

Different curves can reach the pole ($r = 0$) at different values of θ (since a single point, the origin, corresponds to $r = 0$ for ANY θ). Setting $f(\theta) = g(\theta)$ only finds angles where both curves have the SAME r at the SAME θ , so it can miss cases where each curve passes through the origin at a different angle — the pole must always be checked as a separate case.

Area using symmetry to simplify computation

- 14 Explain how the symmetry of $r = 3\cos(2\theta)$ about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.

Since $r = 3\cos(2\theta)$ produces 4 congruent petals by symmetry (reflections across both the x - and y -axes map petals onto each other), computing the area of ONE petal via a single definite integral and multiplying by 4 avoids having to integrate piecewise across the full $[0, 2\pi]$ range while tracking sign changes.

Area review: full worked example

- 15 Find the area enclosed by $r = \sqrt{\theta}$ for $\theta \in [0, \pi]$ (a spiral arc closed off by the segment back to the pole).

$$A = \frac{1}{2} \int_0^{\pi} (\sqrt{\theta})^2 d\theta = \frac{1}{2} \int_0^{\pi} \theta d\theta = \frac{1}{2} \left[\frac{\theta^2}{2} \right]_0^{\pi} = \frac{\pi^2}{4}.$$

- 16 Find the area enclosed by $r = 4\sin \theta$ (a circle) for $\theta \in [0, \pi]$, and verify your answer matches the known area formula for a circle of the appropriate radius.

$$A = \frac{1}{2} \int_0^{\pi} 16\sin^2 \theta d\theta = 8 \int_0^{\pi} \frac{1 - \cos 2\theta}{2} d\theta = 4 \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi} = 4\pi. \text{ This matches } \pi r^2 \text{ for a circle of radius 2 (since } r = 4\sin \theta \text{ traces a circle of radius 2 centered at } (0, 2)\text{): } \pi(2)^2 = 4\pi \checkmark.$$

Comparing polar area to rectangular area (conceptual)

- 17 Explain why the polar area formula $A = \frac{1}{2} \int r^2 d\theta$ uses r^2 rather than just r , by relating it to the area of a thin circular sector.

A thin sector of angle $d\theta$ and radius r has area approximately $\frac{1}{2}r^2 d\theta$ (a fraction $\frac{d\theta}{2\pi}$ of the full circle's area πr^2). Summing (integrating) these thin sector areas over the range of θ gives the total polar area formula — the r^2 arises naturally from the geometry of a circular sector, not from any arbitrary convention.

- 18 A student claims the area between $r = 1$ and $r = 2$ (an annulus, full circle) is simply $\frac{1}{2} \int_0^{2\pi} (2 - 1)^2 d\theta$. Explain the error and give the correct setup.

The error is subtracting the radii before squaring — the correct polar formula requires subtracting the SQUARED radii, not squaring the difference of radii. Correct setup:

$$A = \frac{1}{2} \int_0^{2\pi} (2^2 - 1^2) d\theta = \frac{1}{2} \int_0^{2\pi} 3 d\theta = 3\pi \text{ (matching } \pi(2^2 - 1^2) = 3\pi \text{ from the standard annulus-area formula).}$$

Multiple choice — easy 15

- 19 Find the area enclosed by $r = 1$ for $\theta \in [0, \frac{\pi}{2}]$.

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. $-\frac{7\pi}{4}$

d. $\frac{3\pi}{4}$

$$A = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1^2 d\theta = \frac{1}{2}(1)\left(\frac{\pi}{2}\right) = \frac{\pi}{4}.$$

- 20 Find the area enclosed by $r = 2$ for $\theta \in [0, 2\pi]$.

a. 4π ✓

b. 5π

c. $\frac{9\pi}{2}$

d. 8π

$$A = \frac{1}{2} \int_0^{2\pi} 2^2 d\theta = \frac{1}{2}(4)(2\pi) = 4\pi.$$

- 21 Find the area enclosed by $r = 3$ for $\theta \in [0, \frac{3\pi}{2}]$.

a. $\frac{31\pi}{4}$

b. $\frac{27\pi}{2}$

c. $\frac{27\pi}{4}$ ✓

d. $\frac{29\pi}{4}$

$$A = \frac{1}{2} \int_0^{\frac{3\pi}{2}} 3^2 d\theta = \frac{1}{2}(9)\left(\frac{3\pi}{2}\right) = \frac{27\pi}{4}.$$

22 Find the area enclosed by $r = 7$ for $\theta \in [0, \pi]$.

a. $\frac{49\pi}{2}$ ✓

b. 25π

c. $\frac{53\pi}{2}$

d. 49π

$$A = \frac{1}{2} \int_0^{\pi} 7^2 d\theta = \frac{1}{2}(49)(\pi) = \frac{49\pi}{2}.$$

23 Find the area enclosed by $r = 1$ for $\theta \in [0, \pi]$.

a. $-\frac{3\pi}{2}$

b. π

c. $\frac{5\pi}{2}$

d. $\frac{\pi}{2}$ ✓

$$A = \frac{1}{2} \int_0^{\pi} 1^2 d\theta = \frac{1}{2}(1)(\pi) = \frac{\pi}{2}.$$

24 Find the area enclosed by $r = 6$ for $\theta \in [0, 2\pi]$.

a. $\frac{73\pi}{2}$

b. 36π ✓

c. 37π

d. 72π

$$A = \frac{1}{2} \int_0^{2\pi} 6^2 d\theta = \frac{1}{2}(36)(2\pi) = 36\pi.$$

25 Find the area enclosed by $r = 6$ for $\theta \in [0, \frac{\pi}{2}]$.

a. 18π

b. 7π

c. $\frac{19\pi}{2}$

d. 9π ✓

$$A = \frac{1}{2} \int_0^{\frac{\pi}{2}} 6^2 d\theta = \frac{1}{2}(36)(\frac{\pi}{2}) = 9\pi.$$

26 Find the area enclosed by $r = 6$ for $\theta \in [0, \frac{3\pi}{2}]$.

a. 27π ✓

b. $\frac{55\pi}{2}$

c. 26π

d. 54π

$$A = \frac{1}{2} \int_0^{\frac{3\pi}{2}} 6^2 d\theta = \frac{1}{2}(36)(\frac{3\pi}{2}) = 27\pi.$$

27 Find the area enclosed by $r = 9$ for $\theta \in [0, \frac{\pi}{2}]$.

a. $\frac{81\pi}{4}$ ✓

b. $\frac{81\pi}{2}$

c. $\frac{89\pi}{4}$

d. $\frac{83\pi}{4}$

$$A = \frac{1}{2} \int_0^{\frac{\pi}{2}} 9^2 d\theta = \frac{1}{2}(81)(\frac{\pi}{2}) = \frac{81\pi}{4}.$$

28 Find the area enclosed by $r = 3$ for $\theta \in [0, 2\pi]$.

a. 9π ✓

b. 18π

c. $\frac{19\pi}{2}$

d. 7π

$$A = \frac{1}{2} \int_0^{2\pi} 3^2 d\theta = \frac{1}{2}(9)(2\pi) = 9\pi.$$

29 Find the area enclosed by $r = 1$ for $\theta \in [0, \frac{3\pi}{2}]$.

a. $\frac{5\pi}{4}$

b. $\frac{3\pi}{4}$ ✓

c. $\frac{7\pi}{4}$

d. $\frac{3\pi}{2}$

$$A = \frac{1}{2} \int_0^{\frac{3\pi}{2}} 1^2 d\theta = \frac{1}{2}(1)\left(\frac{3\pi}{2}\right) = \frac{3\pi}{4}.$$

30 Find the area enclosed by $r = 9$ for $\theta \in [0, \pi]$.

a. 81π

b. $\frac{83\pi}{2}$

c. $\frac{81\pi}{2}$ ✓

d. 41π

$$A = \frac{1}{2} \int_0^{\pi} 9^2 d\theta = \frac{1}{2}(81)(\pi) = \frac{81\pi}{2}.$$

31 Find the area enclosed by $r = 8$ for $\theta \in [0, \frac{\pi}{2}]$.

a. $\frac{33\pi}{2}$

b. 17π

c. 16π ✓

d. 32π

$$A = \frac{1}{2} \int_0^{\frac{\pi}{2}} 8^2 d\theta = \frac{1}{2}(64)\left(\frac{\pi}{2}\right) = 16\pi.$$

32 Find the area enclosed by $r = 8$ for $\theta \in [0, 2\pi]$.

a. $\frac{129\pi}{2}$

b. 62π

c. 64π ✓

d. 128π

$$A = \frac{1}{2} \int_0^{2\pi} 8^2 d\theta = \frac{1}{2}(64)(2\pi) = 64\pi.$$

33 Find the area enclosed by $r = 4$ for $\theta \in [0, \frac{3\pi}{2}]$.

a. 11π

b. $\frac{25\pi}{2}$

c. 24π

d. 12π ✓

$$A = \frac{1}{2} \int_0^{\frac{3\pi}{2}} 4^2 d\theta = \frac{1}{2}(16)(\frac{3\pi}{2}) = 12\pi.$$

Multiple choice — medium 15

34 Find the area inside the circle $r = 9$ but outside the circle $r = 4$.

a. -7π

b. 25π

c. 77π

d. 65π ✓

$$A = \frac{1}{2} \int_0^{2\pi} (9^2 - 4^2) d\theta = \frac{1}{2}(65)(2\pi) = 65\pi.$$

35 Find the area inside the circle $r = 13$ but outside the circle $r = 5$.

a. -12π

b. 64π

c. 144π ✓

d. 164π

$$A = \frac{1}{2} \int_0^{2\pi} (13^2 - 5^2) d\theta = \frac{1}{2}(144)(2\pi) = 144\pi.$$

36 Find the area inside the circle $r = 8$ but outside the circle $r = 5$.

a. 39π ✓

b. 59π

c. 9π

d. -17π

$$A = \frac{1}{2} \int_0^{2\pi} (8^2 - 5^2) d\theta = \frac{1}{2}(39)(2\pi) = 39\pi.$$

37 Find the area inside the circle $r = 10$ but outside the circle $r = 6$.

a. -26π

b. 64π ✓

c. 94π

d. 16π

$$A = \frac{1}{2} \int_0^{2\pi} (10^2 - 6^2) d\theta = \frac{1}{2}(64)(2\pi) = 64\pi.$$

38 Find the area inside the circle $r = 4$ but outside the circle $r = 2$.

a. 14π

b. 12π ✓

c. 4π

d. 0π

$$A = \frac{1}{2} \int_0^{2\pi} (4^2 - 2^2) d\theta = \frac{1}{2}(12)(2\pi) = 12\pi.$$

39 Find the area inside the circle $r = 13$ but outside the circle $r = 6$.

a. 163π

b. -23π

c. 49π

d. 133π ✓

$$A = \frac{1}{2} \int_0^{2\pi} (13^2 - 6^2) d\theta = \frac{1}{2}(133)(2\pi) = 133\pi.$$

40 Find the area inside the circle $r = 13$ but outside the circle $r = 7$.

a. 162π

b. -36π

c. 120π ✓

d. 36π

$$A = \frac{1}{2} \int_0^{2\pi} (13^2 - 7^2) d\theta = \frac{1}{2}(120)(2\pi) = 120\pi.$$

41 Find the area inside the circle $r = 10$ but outside the circle $r = 5$.

a. -15π

b. 75π ✓

c. 95π

d. 25π

$$A = \frac{1}{2} \int_0^{2\pi} (10^2 - 5^2) d\theta = \frac{1}{2}(75)(2\pi) = 75\pi.$$

42 Find the area inside the circle $r = 15$ but outside the circle $r = 7$.

a. 64π

b. 176π ✓

c. -34π

d. 218π

$$A = \frac{1}{2} \int_0^{2\pi} (15^2 - 7^2) d\theta = \frac{1}{2}(176)(2\pi) = 176\pi.$$

43 Find the area inside the circle $r = 8$ but outside the circle $r = 3$.

a. 55π ✓

b. 61π

c. 25π

d. $-\pi$

$$A = \frac{1}{2} \int_0^{2\pi} (8^2 - 3^2) d\theta = \frac{1}{2}(55)(2\pi) = 55\pi.$$

44 Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

a. π

b. $-\pi$

c. 5π ✓

d. 7π

$$A = \frac{1}{2} \int_0^{2\pi} (3^2 - 2^2) d\theta = \frac{1}{2}(5)(2\pi) = 5\pi.$$

45 Find the area inside the circle $r = 8$ but outside the circle $r = 1$.

a. 49π

b. 7π

c. 61π

d. 63π ✓

$$A = \frac{1}{2} \int_0^{2\pi} (8^2 - 1^2) d\theta = \frac{1}{2}(63)(2\pi) = 63\pi.$$

46 Find the area inside the circle $r = 10$ but outside the circle $r = 3$.

a. 49π

b. 97π

c. 91π ✓

d. π

$$A = \frac{1}{2} \int_0^{2\pi} (10^2 - 3^2) d\theta = \frac{1}{2}(91)(2\pi) = 91\pi.$$

47 Find the area inside the circle $r = 5$ but outside the circle $r = 4$.

a. 9π ✓

b. π

c. -11π

d. 21π

$$A = \frac{1}{2} \int_0^{2\pi} (5^2 - 4^2) d\theta = \frac{1}{2}(9)(2\pi) = 9\pi.$$

48 Find the area inside the circle $r = 11$ but outside the circle $r = 8$.

a. 9π

b. -53π

c. 113π

d. 57π ✓

$$A = \frac{1}{2} \int_0^{2\pi} (11^2 - 8^2) d\theta = \frac{1}{2}(57)(2\pi) = 57\pi.$$

Multiple choice — hard 15

49 Find the total area enclosed by the curve $r = 8 + 7\sin \theta$.

a. 113π

b. $\frac{177\pi}{2}$ ✓

c. $\frac{179\pi}{2}$

d. $\frac{225\pi}{2}$

Expanding $(8 + 7\sin \theta)^2 = 8^2 + 2(8)(7)\sin \theta + 7^2\sin^2 \theta$ and integrating over $[0, 2\pi]$, the linear $\sin \theta$ term vanishes and $\sin^2 \theta$ averages to $\frac{1}{2}$, leaving $A = \pi(8^2 + \frac{7^2}{2}) = \frac{177\pi}{2}$.

50 Find the total area enclosed by the curve $r = 7 + 6\cos \theta$.

a. 85π

b. 68π

c. $\frac{169\pi}{2}$

d. 67π ✓

Expanding $(7 + 6\cos \theta)^2 = 7^2 + 2(7)(6)\cos \theta + 6^2\cos^2 \theta$ and integrating over $[0, 2\pi]$, the linear $\cos \theta$ term vanishes and $\cos^2 \theta$ averages to $\frac{1}{2}$, leaving $A = \pi(7^2 + \frac{6^2}{2}) = 67\pi$.

51 Find the total area enclosed by the curve $r = 5 + 9\sin \theta$.

a. 98π

b. 106π

c. $\frac{133\pi}{2}$

d. $\frac{131\pi}{2}$ ✓

Expanding $(5 + 9\sin \theta)^2 = 5^2 + 2(5)(9)\sin \theta + 9^2\sin^2 \theta$ and integrating over $[0, 2\pi]$, the linear $\sin \theta$ term vanishes and $\sin^2 \theta$ averages to $\frac{1}{2}$, leaving $A = \pi(5^2 + \frac{9^2}{2}) = \frac{131\pi}{2}$.

52 Find the total area enclosed by the curve $r = 4 + 9\cos \theta$.

- a. $\frac{169\pi}{2}$
- b. 97π
- c. $\frac{115\pi}{2}$
- d. $\frac{113\pi}{2}$ ✓

Expanding $(4 + 9\cos \theta)^2 = 4^2 + 2(4)(9)\cos \theta + 9^2\cos^2 \theta$ and integrating over $[0, 2\pi]$, the linear $\cos \theta$ term vanishes and $\cos^2 \theta$ averages to $\frac{1}{2}$, leaving $A = \pi(4^2 + \frac{9^2}{2}) = \frac{113\pi}{2}$.

53 Find the total area enclosed by the curve $r = 3 + 5\cos \theta$.

- a. 34π
- b. 32π
- c. $\frac{45\pi}{2}$
- d. $\frac{43\pi}{2}$ ✓

Expanding $(3 + 5\cos \theta)^2 = 3^2 + 2(3)(5)\cos \theta + 5^2\cos^2 \theta$ and integrating over $[0, 2\pi]$, the linear $\cos \theta$ term vanishes and $\cos^2 \theta$ averages to $\frac{1}{2}$, leaving $A = \pi(3^2 + \frac{5^2}{2}) = \frac{43\pi}{2}$.

54 Find the area of one petal of $r = 4\cos(2\theta)$.

- a. $\frac{9\pi}{4}$
- b. 2π ✓
- c. 4π
- d. 8π

Using $A_{\text{petal}} = \frac{k^2\pi}{4n}$ with $k = 4$, $n = 2$: $A = \frac{16\pi}{8} = 2\pi$.

55 Find the area of one petal of $r = 4\sin(7\theta)$.

- a. $\frac{23\pi}{28}$
- b. $\frac{8\pi}{7}$
- c. $\frac{16\pi}{7}$
- d. $\frac{4\pi}{7}$ ✓

Using $A_{\text{petal}} = \frac{k^2\pi}{4n}$ with $k = 4$, $n = 7$: $A = \frac{16\pi}{28} = \frac{4\pi}{7}$.

56 Find the area of one petal of $r = 2\sin(5\theta)$.

- a. $\frac{4\pi}{5}$
- b. $\frac{7\pi}{10}$
- c. $\frac{2\pi}{5}$
- d. $\frac{\pi}{5}$ ✓

Using $A_{\text{petal}} = \frac{k^2\pi}{4n}$ with $k = 2$, $n = 5$: $A = \frac{4\pi}{20} = \frac{\pi}{5}$.

57 Find the area of one petal of $r = 3\cos(6\theta)$.

- a. $\frac{3\pi}{2}$
- b. $\frac{3\pi}{8}$ ✓
- c. $\frac{3\pi}{4}$
- d. $\frac{7\pi}{8}$

Using $A_{\text{petal}} = \frac{k^2\pi}{4n}$ with $k = 3$, $n = 6$: $A = \frac{9\pi}{24} = \frac{3\pi}{8}$.

58 Find the area of one petal of $r = 4\sin(2\theta)$.

- a. $\frac{5\pi}{2}$
- b. 8π
- c. 4π
- d. 2π ✓

Using $A_{\text{petal}} = \frac{k^2\pi}{4n}$ with $k = 4$, $n = 2$: $A = \frac{16\pi}{8} = 2\pi$.

59 Find the area inside the circle $r = 9$ but outside the circle $r = 2$, for $\theta \in [0, \pi]$ only (half the full region).

a. 77π

b. $\frac{79\pi}{2}$

c. $\frac{77\pi}{2}$ ✓

d. $\frac{49\pi}{2}$

$$A = \frac{1}{2} \int_0^{\pi} (9^2 - 2^2) d\theta = \frac{1}{2} (77) (\pi) = \frac{77\pi}{2}.$$

60 Find the area inside the circle $r = 11$ but outside the circle $r = 7$, for $\theta \in [0, \pi]$ only (half the full region).

a. 8π

b. 72π

c. 36π ✓

d. 37π

$$A = \frac{1}{2} \int_0^{\pi} (11^2 - 7^2) d\theta = \frac{1}{2} (72) (\pi) = 36\pi.$$

61 Find the area inside the circle $r = 3$ but outside the circle $r = 2$, for $\theta \in [0, \pi]$ only (half the full region).

a. $\frac{7\pi}{2}$

b. $\frac{5\pi}{2}$ ✓

c. $\frac{\pi}{2}$

d. 5π

$$A = \frac{1}{2} \int_0^{\pi} (3^2 - 2^2) d\theta = \frac{1}{2} (5) (\pi) = \frac{5\pi}{2}.$$

62 Find the area inside the circle $r = 3$ but outside the circle $r = 1$, for $\theta \in [0, \pi]$ only (half the full region).

a. 5π

b. 8π

c. 4π ✓

d. 2π

$$A = \frac{1}{2} \int_0^{\pi} (3^2 - 1^2) d\theta = \frac{1}{2}(8)(\pi) = 4\pi.$$

63 Find the area inside the circle $r = 13$ but outside the circle $r = 5$, for $\theta \in [0, \pi]$ only (half the full region).

a. 72π ✓

b. 73π

c. 32π

d. 144π

$$A = \frac{1}{2} \int_0^{\pi} (13^2 - 5^2) d\theta = \frac{1}{2}(144)(\pi) = 72\pi.$$